DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 4: Assignment

CAPM · Beta & the SML · Beta Estimation

Instructions

- This assignment covers Lectures 10–12 and the Week 4 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script or notebook cell; submit your code alongside your written answers.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (15 marks)

A1. **(4 marks)** Explain, in 2–3 sentences, the equilibrium argument for why the tangency portfolio from Lecture 9 must equal the market portfolio in CAPM.

A2. **(4 marks)** Explain why unsystematic risk earns no reward under CAPM, using the idea of a "free lunch" that would otherwise exist.

A3. **(3 marks)** State one assumption underlying CAPM, and explain why it can be unrealistic yet the model still be useful.

A4. **(4 marks)** True or false, with justification: "The Capital Market Line and the Security Market Line are really the same line, just plotted with different axis labels."

Part B – CAPM Calculations (25 marks)

B1. **(6 marks)** $r_f = 6\%$, $\mathbb{E}[R_m] = 15\%$. Compute $\mathbb{E}[R_i]$ under CAPM for $\beta_i = 1.4,\ 0.5,\ 0$.

B2. **(6 marks)** A stock has $\beta = 1.5$ and an analyst's actual expected return estimate of 18%. Using $r_f = 6\%, \mathbb{E}[R_m] = 15\%$, compute the stock's alpha. Is it over- or under-priced according to CAPM?

B3. **(6 marks)** A stock has $\mathrm{Cov}(R_i, R_m) = 180$ and $\mathrm{Var}(R_m) = 150$ (both in $\%^2$). Find β_i. If $r_f = 5\%, \mathbb{E}[R_m] = 12\%$, find $\mathbb{E}[R_i]$.

B4. **(7 marks)** A portfolio holds stock X ($\beta_X = 1.2$, weight 0.5) and stock Y ($\beta_Y = 0.8$, weight 0.5). Compute the portfolio beta, and its CAPM expected return using $r_f = 5\%, \mathbb{E}[R_m] = 13\%$.

Part C – Beta Interpretation and Portfolio Beta (20 marks)

C1. (5 marks) Classify each as aggressive, defensive, market-neutral, or a rare negative-beta asset: $\beta = 1.6$; $\beta = 0.4$; $\beta = 0$; $\beta = -0.2$.

C2. (5 marks) Explain why an asset with a negative beta could be valuable to hold in a portfolio, even if its own expected return (per CAPM) is below the risk-free rate.

C3. (10 marks) An investor's portfolio is currently fully invested in risky assets with a portfolio beta of $\beta_p = 1.3$. They want to reduce their portfolio beta to 0.8 by shifting some fraction of wealth into the risk-free asset (which has $\beta = 0$), keeping the relative proportions among the risky assets unchanged. What fraction of the portfolio should be shifted into the risk-free asset? (Hint: if y is the fraction remaining in the risky mix, the new portfolio beta is $y\beta_p$.)

Part D – Beta Estimation (20 marks)

D1. (8 marks) The table below gives six periods of market and stock returns (%):

Period	1	2	3	4	5	6
R_m	2	5	-3	4	-1	3
R_i	3	7	-5	6	-2	5

Compute $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R_i, R_m)/\widehat{\mathrm{Var}}(R_m)$ by hand, showing the sample means, deviations, and every cross-product used.

D2. (4 marks) A regression of a stock's returns on the market gives $R^2 = 0.49$. What fraction of this stock's risk is unsystematic? What is the implied $|\rho_{i,m}|$?

D3. (4 marks) Explain, in your own words, Roll's critique of CAPM.

D4. (4 marks) A company recently underwent a major merger that significantly changed its business mix. Explain why its *rolling* beta (estimated on, say, the trailing 24 months) might currently give a more useful estimate of its true systematic risk than its *full-sample* beta computed over the last 10 years.

Part E – Computational (15 marks)

E1. (15 marks) [Python] Using your Lab code (or a fresh, self-contained script) and the bundled `sample_market_stocks.csv` data, compute rolling beta for `STOCK_D` using *two* different window lengths: 30 days and 90 days. Plot both rolling beta series on the same chart. In 3–4 sentences, comment on the trade-off between window length and (a) how noisy/jumpy the estimate looks, and (b) how quickly it responds to a genuine change in the stock's systematic risk.

Part F – Integrative Question (5 marks)

F1. (5 marks) In Week 3, pricing a portfolio's risk required knowing the *entire* covariance matrix Σ – every asset's covariance with every other asset. Explain why, under CAPM, an investor evaluating a single stock's contribution to systematic risk only needs that stock's beta relative to the market, rather than its covariance with every other individual holding. What key CAPM assumption or result makes this simplification valid?

End of Assignment.